

Totally Integrated Automation Portal

GRAFCET [FB1]

GRAFCET Properties

General

Name	GRAFCET	Number	1	Type	FB
Language	GRAPH	Numbering	Automatic	Network language	LAD
Block version	V2.0				

Information

Title		Author		Comment	
Family		Version	0.1	User-defined ID	

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HM I/O PC UA/Web API	Visible in HMI engineering	Set-point	Supervision	Comment
▼ Input									
OFF_SQ	Bool	false	Non-retain	False	False	False	False		Turn sequence off
INIT_SQ	Bool	false	Non-retain	False	False	False	False		Set sequence to initial state
ACK_EF	Bool	false	Non-retain	False	False	False	False		Acknowledge all errors and faults
S_PREV	Bool	false	Non-retain	False	False	False	False		Output previous step in parameter S_NO
S_NEXT	Bool	false	Non-retain	False	False	False	False		Indicate next step in parameter S_NO
SW_AUTO	Bool	false	Non-retain	False	False	False	False		Automatic mode
SW_TAP	Bool	false	Non-retain	False	False	False	False		Semiautomatic/switch with transition
SW_TOP	Bool	false	Non-retain	False	False	False	False		Semiautomatic/ignore transition
SW_MAN	Bool	false	Non-retain	False	False	False	False		Manual mode
S_SEL	Int	0	Non-retain	False	False	False	False		Select step to be output to S_NO
S_ON	Bool	false	Non-retain	False	False	False	False		Activate step indicated in S_NO
S_OFF	Bool	false	Non-retain	False	False	False	False		Disable step indicated in S_NO
T_PUSH	Bool	false	Non-retain	False	False	False	False		Enable transition to switch in semi automatic mode
▼ Output									
S_NO	Int	0	Non-retain	False	False	False	False		Step number

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
S_MORE	Bool	false	Non-retain	False	False	False	False			More steps are available and can be shown in S_NO
S_ACTIVE	Bool	false	Non-retain	False	False	False	False			Step indicated in S_NO is active
ERR_FLT	Bool	false	Non-retain	False	False	False	False			Interlock or supervision group error
AUTO_ON	Bool	false	Non-retain	False	False	False	False			Automatic mode is active
TAP_ON	Bool	false	Non-retain	False	False	False	False			Semiautomatic mode/step with transition enabled
TOP_ON	Bool	false	Non-retain	False	False	False	False			Semiautomatic mode/ignore transition enabled
MAN_ON	Bool	false	Non-retain	False	False	False	False			Manual mode is active
InOut										
▼ Static										
▼ RT_DATA	G7_RTDataPlus_V2		Non-retain	False	False	False	True			Internal data area
S_DISPLAY	Int	0	Non-retain	False	False	False	False			Internal display of output parameter S_NO
S_SEL_OLD	Int	0	Non-retain	False	False	False	False			Previous value in S_SEL
S_DISPIDX	USInt	255	Non-retain	False	False	False	False			Index of the step in S_NO
T_DISPIDX	USInt	255	Non-retain	False	False	False	False			Index of the transition displayed in T_NO
▼ MOP_EDGE	G7_MOP-Plus_V2		Non-retain	False	False	False	True			Mode in last cycle
AUTO	Bool	false	Non-retain	False	False	False	False			Status: automatic mode
MAN	Bool	false	Non-retain	False	False	False	False			Status: manual mode
TAP	Bool	false	Non-retain	False	False	False	False			Status: semi automatic/switch with transition
TOP	Bool	false	Non-retain	False	False	False	False			Status: semi automatic/ignore transition
ACK_S	Bool	false	Non-retain	False	False	False	False			Request: acknowledge step at parameter S_NO
REG_S	Bool	false	Non-retain	False	False	False	False			Request: register step indicated in S_NO

Totally Integrated Automation Portal										
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T_PREV	Bool	false	Non-retain	False	False	False	False			Request: output previous valid transition in T_NO
T_NEXT	Bool	false	Non-retain	False	False	False	False			Request: output next valid transition in T_NO
LOCK	Bool	false	Non-retain	False	False	False	False			Status: interlocks activated
SUP	Bool	false	Non-retain	False	False	False	False			Status: supervisions activated
ACKREQ	Bool	false	Non-retain	False	False	False	False			Status: acknowledgment required
SSKIP	Bool	false	Non-retain	False	False	False	False			Status: "Skip steps" enabled
OFF	Bool	false	Non-retain	False	False	False	False			Request: deactivate all steps
INIT	Bool	false	Non-retain	False	False	False	False			Request: set sequence to initial state
HALT	Bool	false	Non-retain	False	False	False	False			Status: sequence halted
TMS_HALT	Bool	false	Non-retain	False	False	False	False			Status: all internal timers held
OPS_ZERO	Bool	false	Non-retain	False	False	False	False			Status: set all operands processed with N, L, D instructions to 0
SACT_DISP	Bool	false	Non-retain	False	False	False	False			Status: display active steps only
SEF_DISP	Bool	false	Non-retain	False	False	False	False			Status: display only steps with errors and disrupted steps
SALL_DISP	Bool	false	Non-retain	False	False	False	False			Status: display all steps
S_PREV	Bool	false	Non-retain	False	False	False	False			Request: output previous step to S_NO
S_NEXT	Bool	false	Non-retain	False	False	False	False			Request: Output next step at S_NO parameter
S_SELOK	Bool	false	Non-retain	False	False	False	False			Request: output step number from S_SEL to S_NO
S_ON	Bool	false	Non-retain	False	False	False	False			Request: activate step indicated in S_NO
S_OFF	Bool	false	Non-retain	False	False	False	False			Request: deactivate step at parameter S_NO

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T_PUSH	Bool	false	Non-retain	False	False	False	False			Request: transition switching enabled
REG	Bool	false	Non-retain	False	False	False	False			Request: register all interlock and supervision errors
ACK	Bool	false	Non-retain	False	False	False	False			Request: acknowledge all interlock and supervision errors
IL_PERM	Bool	false	Non-retain	False	False	False	False			Status: permanent processing of all interlocks
T_PERM	Bool	false	Non-retain	False	False	False	False			Status: permanent processing of all transitions
ILP_MAN	Bool	false	Non-retain	False	False	False	False			Status: permanent processing of all interlocks in manual mode
LMODE	Bool	false	Non-retain	False	False	False	False			Status: learning mode is active
▼ MOP	G7_MOP-Plus_V2		Non-retain	False	False	False	True			Mode
AUTO	Bool	true	Non-retain	False	False	False	False			Status: automatic mode
MAN	Bool	false	Non-retain	False	False	False	False			Status: manual mode
TAP	Bool	false	Non-retain	False	False	False	False			Status: semi automatic/switch with transition
TOP	Bool	false	Non-retain	False	False	False	False			Status: semi automatic/ignore transition
ACK_S	Bool	false	Non-retain	False	False	False	False			Request: acknowledge step at parameter S_NO
REG_S	Bool	false	Non-retain	False	False	False	False			Request: register step indicated in S_NO
T_PREV	Bool	false	Non-retain	False	False	False	False			Request: output previous valid transition in T_NO
T_NEXT	Bool	false	Non-retain	False	False	False	False			Request: output next valid transition in T_NO
LOCK	Bool	true	Non-retain	False	False	False	False			Status: interlocks activated
SUP	Bool	true	Non-retain	False	False	False	False			Status: supervisions activated
ACKREQ	Bool	true	Non-retain	False	False	False	False			Status: acknowledgment required

Totally Integrated Automation Portal										
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SSKIP	Bool	false	Non-retain	False	False	False	False			Status: "Skip steps" enabled
OFF	Bool	false	Non-retain	False	False	False	False			Request: deactivate all steps
INIT	Bool	true	Non-retain	False	False	False	False			Request: set sequence to initial state
HALT	Bool	false	Non-retain	False	False	False	False			Status: sequence halted
TMS_HALT	Bool	false	Non-retain	False	False	False	False			Status: all internal timers held
OPS_ZERO	Bool	false	Non-retain	False	False	False	False			Status: set all operands processed with N, L, D instructions to 0
SACT_DISP	Bool	true	Non-retain	False	False	False	False			Status: display active steps only
SEF_DISP	Bool	false	Non-retain	False	False	False	False			Status: display only steps with errors and disrupted steps
SALL_DISP	Bool	false	Non-retain	False	False	False	False			Status: display all steps
S_PREV	Bool	false	Non-retain	False	False	False	False			Request: output previous step to S_NO
S_NEXT	Bool	false	Non-retain	False	False	False	False			Request: Output next step at S_NO parameter
S_SELOK	Bool	false	Non-retain	False	False	False	False			Request: output step number from S_SEL to S_NO
S_ON	Bool	false	Non-retain	False	False	False	False			Request: activate step indicated in S_NO
S_OFF	Bool	false	Non-retain	False	False	False	False			Request: deactivate step at parameter S_NO
T_PUSH	Bool	false	Non-retain	False	False	False	False			Request: transition switching enabled
REG	Bool	false	Non-retain	False	False	False	False			Request: register all interlock and supervision errors
ACK	Bool	false	Non-retain	False	False	False	False			Request: acknowledge all interlock and supervision errors
IL_PERM	Bool	false	Non-retain	False	False	False	False			Status: permanent processing of all interlocks

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Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
T_PERM	Bool	false	Non-retain	False	False	False	False		Status: permanent processing of all transitions	
ILP_MAN	Bool	false	Non-retain	False	False	False	False		Status: permanent processing of all interlocks in manual mode	
LMODE	Bool	false	Non-retain	False	False	False	False		Status: learning mode is acitve	
TIME_DELTA	Time	T#0ms	Non-retain	False	False	False	False		Cycle time	
▼ SQ_FLAGS	G7_SQFlagsPlus_V2		Non-retain	False	False	False	True		Sequence bit memory	
ERR_FLT	Bool	false	Non-retain	False	False	False	False		Interlock and supervision group error	
ERROR	Bool	false	Non-retain	False	False	False	False		Interlock group error	
FAULT	Bool	false	Non-retain	False	False	False	False		Supervision group error	
RT_FAIL	Bool	false	Non-retain	False	False	False	False		Runtime error	
NO_SNO	Bool	false	Non-retain	False	False	False	False		Requested step number not found	
NF_OFL	Bool	false	Non-retain	False	False	False	False		Overflow: too many ON or OFF requests	
SA_OFL	Bool	false	Non-retain	False	False	False	False		Overflow: too many steps active	
TV_OFL	Bool	false	Non-retain	False	False	False	False		Overflow: too many valid transitions	
MSG_OFL	Bool	false	Non-retain	False	False	False	False		Overflow: not enough system resources for ALARM_S	
NO_SWI	Bool	false	Non-retain	False	False	False	False		Do not switch in this cycle	
CYC_OP	Bool	false	Non-retain	False	False	False	False		Cyclic execution of the sequence after initialization	
AS_MSG	Bool	true	Non-retain	False	False	False	False		Alarms during runtime enabled or disabled by instruction	
AS_SEND	Bool	false	Non-retain	False	False	False	False		Send alarms from WR_USMSG or only enter in diagnostics buffer	

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Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Set-point	Supervision		Comment
SQ_BUSY	Bool	false	Non-retain	False	False	False	False			Internal edge memory bit for sequence processing
SA_BUSY	Bool	false	Non-retain	False	False	False	False			Internal edge memory bit for sequence processing
AS_SIG	Bool	false	Non-retain	False	False	False	False			Edge memory bit for alarms from Alarm_S and Alarm_SQ
PRE_CNT	USInt	1	Non-retain	False	False	False	False			Number of permanent instructions preceding the sequencer
POST_CNT	USInt	1	Non-retain	False	False	False	False			Number of permanent instructions after the sequencer
SQ_CNT	USInt	1	Non-retain	False	False	False	False			Number of branch paths
S_CNT	USInt	22	Non-retain	False	False	False	False			Number of steps
LOCK_CNT	USInt	0	Non-retain	False	False	False	False			Number of interlocks
SUP_CNT	USInt	0	Non-retain	False	False	False	False			Number of supervisions
T_CNT	USInt	23	Non-retain	False	False	False	False			Number of transitions
SQ_PART_CNT	USInt	3	Non-retain	False	False	False	False			Number of branches
MAX_TVAL	USInt	3	Non-retain	False	False	False	False			Max. number of simultaneously valid transitions
MAX_SACT	USInt	1	Non-retain	False	False	False	False			Max. number of simultaneously active steps
AS_MSG	Byte	16#65	Non-retain	False	False	False	False			Alarm flags
▼ EXEC_BITS	Array[0..249] of Bool		Non-retain	False	False	False	False			System-internal
EXEC_BITS[0]	Bool	false	Non-retain	False	False	False	False			System-internal
EXEC_BITS[1]	Bool	false	Non-retain	False	False	False	False			System-internal
EXEC_BITS[2]	Bool	false	Non-retain	False	False	False	False			System-internal
EXEC_BITS[3]	Bool	false	Non-retain	False	False	False	False			System-internal

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EX-EC_BITS[4]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[5]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[6]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[7]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[8]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[9]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[10]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[11]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[12]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[13]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[14]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[15]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[16]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[17]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[18]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[19]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[20]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[21]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[22]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[23]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[24]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[25]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[26]	Bool	false	Non-retain	False	Fals e	False	False				System-internal
EX-EC_BITS[27]	Bool	false	Non-retain	False	Fals e	False	False				System-internal

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EX-EC_BITS[28]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[29]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[30]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[31]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[32]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[33]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[34]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[35]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[36]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[37]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[38]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[39]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[40]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[41]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[42]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[43]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[44]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[45]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[46]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[47]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[48]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[49]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[50]	Bool	false	Non-retain	False	False	False	False			System-internal	
EX-EC_BITS[51]	Bool	false	Non-retain	False	False	False	False			System-internal	

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EX-EC_BITS[52]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[53]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[54]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[55]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[56]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[57]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[58]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[59]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[60]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[61]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[62]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[63]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[64]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[65]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[66]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[67]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[68]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[69]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[70]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[71]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[72]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[73]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[74]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX-EC_BITS[75]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	

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EX-EC_BITS[76]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[77]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[78]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[79]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[80]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[81]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[82]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[83]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[84]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[85]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[86]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[87]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[88]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[89]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[90]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[91]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[92]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[93]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[94]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[95]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[96]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[97]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[98]	Bool	false	Non-retain	False	False	False	False				System-internal
EX-EC_BITS[99]	Bool	false	Non-retain	False	False	False	False				System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
EX- EC_BITS[100]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[101]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[102]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[103]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[104]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[105]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[106]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[107]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[108]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[109]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[110]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[111]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[112]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[113]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[114]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[115]	Bool	false	Non-retain	False	Fals e	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
EX- EC_BITS[116]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[117]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[118]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[119]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[120]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[121]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[122]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[123]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[124]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[125]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[126]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[127]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[128]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[129]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[130]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[131]	Bool	false	Non-retain	False	Fals e	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
EX- EC_BITS[132]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[133]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[134]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[135]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[136]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[137]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[138]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[139]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[140]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[141]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[142]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[143]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[144]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[145]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[146]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[147]	Bool	false	Non-retain	False	Fals e	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
EX- EC_BITS[148]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[149]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[150]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[151]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[152]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[153]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[154]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[155]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[156]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[157]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[158]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[159]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[160]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[161]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[162]	Bool	false	Non-retain	False	False	False	False			System-internal
EX- EC_BITS[163]	Bool	false	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment	
EX- EC_BITS[164]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[165]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[166]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[167]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[168]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[169]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[170]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[171]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[172]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[173]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[174]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[175]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[176]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[177]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[178]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[179]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Set-point	Supervision	Comment	
EX-EC_BITS[180]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[181]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[182]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[183]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[184]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[185]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[186]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[187]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[188]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[189]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[190]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[191]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[192]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[193]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[194]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[195]	Bool	false	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment	
EX- EC_BITS[196]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[197]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[198]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[199]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[200]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[201]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[202]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[203]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[204]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[205]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[206]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[207]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[208]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[209]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[210]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[211]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access-ible from HMI/OPC UA/Web API	Wri-ta-ble from HM I/O PC UA/ Web API	Visible in HMI engi-neering	Set-point	Su-per-vi-sion		Comment
EX-EC_BITS[212]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[213]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[214]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[215]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[216]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[217]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[218]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[219]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[220]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[221]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[222]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[223]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[224]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[225]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[226]	Bool	false	Non-retain	False	False	False	False			System-internal
EX-EC_BITS[227]	Bool	false	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
EX- EC_BITS[228]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[229]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[230]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[231]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[232]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[233]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[234]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[235]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[236]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[237]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[238]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[239]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[240]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[241]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[242]	Bool	false	Non-retain	False	Fals e	False	False			System-internal
EX- EC_BITS[243]	Bool	false	Non-retain	False	Fals e	False	False			System-internal

Totally Integrated Automation Portal											
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment	
EX- EC_BITS[244]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[245]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[246]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[247]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[248]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
EX- EC_BITS[249]	Bool	false	Non-retain	False	Fals e	False	False			System-internal	
▼ OFFSETS	G7_Off- sets- Plus_V2		Non-retain	False	Fals e	False	True			Internal offsets	
SINI_OFFSET	UInt	0	Non-retain	False	Fals e	False	False			Offset of internal array SINI[]	
LSTT_OFF- SET	UInt	2	Non-retain	False	Fals e	False	False			Offset of internal array LSTT[]	
ATAJ_OFF- SET	UInt	25	Non-retain	False	Fals e	False	False			Offset of internal array ATAJ[]	
ATAB_OFF- SET	UInt	48	Non-retain	False	Fals e	False	False			Offset of internal array ATAB[]	
PSTT_OFF- SET	UInt	71	Non-retain	False	Fals e	False	False			Offset of internal array PSTT[]	
NSTT_OFF- SET	UInt	94	Non-retain	False	Fals e	False	False			Offset of internal array NSTT[]	
ASSJ_OFF- SET	UInt	117	Non-retain	False	Fals e	False	False			Offset of internal array ASSJ[]	
ASSB_OFF- SET	UInt	139	Non-retain	False	Fals e	False	False			Offset of internal array ASSB[]	
PTTS_OFF- SET	UInt	161	Non-retain	False	Fals e	False	False			Offset of internal array PTTS[]	
NTTS_OFF- SET	UInt	183	Non-retain	False	Fals e	False	False			Offset of internal array NTTS[]	
SW_SQTS_O FFSET	UInt	205	Non-retain	False	Fals e	False	False			Offset of internal array SW_SQTS[]	
SWITCH_OFF SET	UInt	227	Non-retain	False	Fals e	False	False			Offset of internal array SWITCH[]	
TVX_OFFSET	UInt	229	Non-retain	False	Fals e	False	False			Offset of internal array TVX[]	
TTX_OFFSET	UInt	233	Non-retain	False	Fals e	False	False			Offset of internal array TTX[]	

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
TSX_OFFSET	UInt	237	Non-retain	False	False	False	False			Offset of internal array TSX[]
S00X_OFF- SET	UInt	239	Non-retain	False	False	False	False			Offset of internal array S00X[]
SOFFX_OFF- SET	UInt	241	Non-retain	False	False	False	False			Offset of internal array SOFFX[]
SONX_OFF- SET	UInt	243	Non-retain	False	False	False	False			Offset of internal array SONX[]
SAX_OFFSET	UInt	245	Non-retain	False	False	False	False			Offset of internal array SAX[]
SERRX_OFF- SET	UInt	247	Non-retain	False	False	False	False			Offset of internal array SERRX[]
SMX_OFF- SET	UInt	270	Non-retain	False	False	False	False			Offset of internal array SMX[]
SOX_OFFSET	UInt	293	Non-retain	False	False	False	False			Offset of internal array SOX[]
S1X_OFFSET	UInt	316	Non-retain	False	False	False	False			Offset of internal array S1X[]
THRESH- OLD_SUP	USInt	0	Non-retain	False	False	False	False			Threshold for step activation time
THRESH- OLD_WARN	USInt	0	Non-retain	False	False	False	False			Threshold for step activation time (warning only)
▼ Trans1	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	1	Non-retain	False	False	False	False			Indicates the user- defined transition number
▼ Trans2	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	2	Non-retain	False	False	False	False			Indicates the user- defined transition number
▼ Trans3	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	3	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans4	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	4	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans5	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	5	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans6	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	6	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans7	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	7	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans8	G7_Transition-Plus_V2		Non-retain	False	False	False	True			Transition structure
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	8	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans9	G7_Transition-Plus_V2		Non-retain	False	False	False	True			Transition structure
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	9	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans10	G7_Transition-Plus_V2		Non-retain	False	False	False	True			Transition structure
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	10	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans11	G7_Transition-Plus_V2		Non-retain	False	False	False	True			Transition structure
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satisfied

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	11	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans12	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	12	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans13	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	13	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans14	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	14	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans15	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
TNO	Int	15	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans16	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	16	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans17	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	17	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans18	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	18	Non-retain	False	False	False	False			Indicates the user-defined transition number
▼ Trans19	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	19	Non-retain	False	False	False	False			Indicates the user-defined transition number

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HMI I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
▼ Trans20	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	20	Non-retain	False	False	False	False			Indicates the user- defined transition number
▼ Trans21	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	21	Non-retain	False	False	False	False			Indicates the user- defined transition number
▼ Trans22	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	22	Non-retain	False	False	False	False			Indicates the user- defined transition number
▼ Trans23	G7_Transi- tion- Plus_V2		Non-retain	False	False	False	True			Transition struc- ture
TV	Bool	false	Non-retain	False	False	False	False			Transition is valid
TT	Bool	false	Non-retain	False	False	False	False			Transition is satis- fied
TS	Bool	false	Non-retain	False	False	False	False			Transition switches
TNO	Int	23	Non-retain	False	False	False	False			Indicates the user- defined transition number
▼ Reset/Reposo	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	1	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False		System-internal	
▼ A+	G7_Step-Plus_V2		Non-retain	False	False	False	True		Step structure	
S1	Bool	false	Non-retain	False	False	False	False		Step is activated	
L1	Bool	false	Non-retain	False	False	False	False		interlock leaving state	
V1	Bool	false	Non-retain	False	False	False	False		Supervision entering state	
R1	Bool	false	Non-retain	False	False	False	False		Reserved	
A1	Bool	false	Non-retain	False	False	False	False		Error is acknowledged	
S0	Bool	false	Non-retain	False	False	False	False		Step is deactivated	
L0	Bool	false	Non-retain	False	False	False	False		Interlock entering state	
V0	Bool	false	Non-retain	False	False	False	False		Supervision leaving state	
X	Bool	false	Non-retain	False	False	False	False		Step is active	
LA	Bool	false	Non-retain	False	False	False	False		Interlock is not satisfied	
VA	Bool	false	Non-retain	False	False	False	False		Supervision active	
RA	Bool	false	Non-retain	False	False	False	False		Reserved	
AA	Bool	false	Non-retain	False	False	False	False		Reserved	
SS	Bool	false	Non-retain	False	False	False	False		System-internal	
LS	Bool	false	Non-retain	False	False	False	False		Direct result of the programmed interlock	
VS	Bool	false	Non-retain	False	False	False	False		Direct result of the programmed supervision	
SNO	Int	2	Non-retain	False	False	False	False		User step number	
T	Time	T#0ms	Non-retain	False	False	False	False		Total step activation time	
U	Time	T#0ms	Non-retain	False	False	False	False		Step activation time without disturbance	
T_MAX	Time	T#10S	Non-retain	False	False	False	False		Maximal step activation time	
T_WARN	Time	T#7S	Non-retain	False	False	False	False		Warning time	

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ A-	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	3	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ B+	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	4	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ C+	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
SNO	Int	5	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ B-	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	6	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ D sale	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	7	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ D-	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	8	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ C-	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	9	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ E+,F+	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	10	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ E-,F-	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion	Comment	
LO	Bool	false	Non-retain	False	False	False	False			Interlock entering state
VO	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	11	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ A++	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	12	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ A--	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	13	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ B++	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	14	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble from HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ C++	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	15	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ B--	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	16	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ D_out	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	17	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/Web API	Wri- ta- ble from HM I/O PC UA/ Web API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ D--	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
SNO	Int	18	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ C --	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neer- ing	Set- point	Su- per- vi- sion		Comment
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	19	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ E,F++	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion	Comment	
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	20	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ E,F--	G7_Step-Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision entering state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowledged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leaving state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not satisfied
VA	Bool	false	Non-retain	False	False	False	False			Supervision active
RA	Bool	false	Non-retain	False	False	False	False			Reserved

Totally Integrated Automation Portal										
Name	Data type	Default value	Retain	Access- ible from HMI/OP C UA/We b API	Wri- ta- ble fro m HM I/O PC UA/ We b API	Visible in HMI engi- neering	Set- point	Su- per- vi- sion		Comment
AA	Bool	false	Non-retain	False	False	False	False			Reserved
SS	Bool	false	Non-retain	False	False	False	False			System-internal
LS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed inter- lock
VS	Bool	false	Non-retain	False	False	False	False			Direct result of the programmed supervision
SNO	Int	21	Non-retain	False	False	False	False			User step number
T	Time	T#0ms	Non-retain	False	False	False	False			Total step activa- tion time
U	Time	T#0ms	Non-retain	False	False	False	False			Step activation time without dis- turbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False			Maximal step acti- vation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False			Warning time
SM	Bool	false	Non-retain	False	False	False	False			System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False			System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False			System-internal
▼ Cilindros -	G7_Step- Plus_V2		Non-retain	False	False	False	True			Step structure
S1	Bool	false	Non-retain	False	False	False	False			Step is activated
L1	Bool	false	Non-retain	False	False	False	False			interlock leaving state
V1	Bool	false	Non-retain	False	False	False	False			Supervision enter- ing state
R1	Bool	false	Non-retain	False	False	False	False			Reserved
A1	Bool	false	Non-retain	False	False	False	False			Error is acknowl- edged
S0	Bool	false	Non-retain	False	False	False	False			Step is deactivated
L0	Bool	false	Non-retain	False	False	False	False			Interlock entering state
V0	Bool	false	Non-retain	False	False	False	False			Supervision leav- ing state
X	Bool	false	Non-retain	False	False	False	False			Step is active
LA	Bool	false	Non-retain	False	False	False	False			Interlock is not sat- isfied

Totally Integrated Automation Portal

Name	Data type	Default value	Retain	Accessible from HMI/OPC UA/Web API	Writable from HMI/OPC UA/Web API	Visible in HMI engineering	Set-point	Supervision	Comment
VA	Bool	false	Non-retain	False	False	False	False		Supervision active
RA	Bool	false	Non-retain	False	False	False	False		Reserved
AA	Bool	false	Non-retain	False	False	False	False		Reserved
SS	Bool	false	Non-retain	False	False	False	False		System-internal
LS	Bool	false	Non-retain	False	False	False	False		Direct result of the programmed interlock
VS	Bool	false	Non-retain	False	False	False	False		Direct result of the programmed supervision
SNO	Int	22	Non-retain	False	False	False	False		User step number
T	Time	T#0ms	Non-retain	False	False	False	False		Total step activation time
U	Time	T#0ms	Non-retain	False	False	False	False		Step activation time without disturbance
T_MAX	Time	T#10S	Non-retain	False	False	False	False		Maximal step activation time
T_WARN	Time	T#7S	Non-retain	False	False	False	False		Warning time
SM	Bool	false	Non-retain	False	False	False	False		System-internal
H_IL_ERR	Byte	16#0	Non-retain	False	False	False	False		System-internal
H_SV_FLT	Byte	16#04	Non-retain	False	False	False	False		System-internal
Temp									
Constant									

Alarms

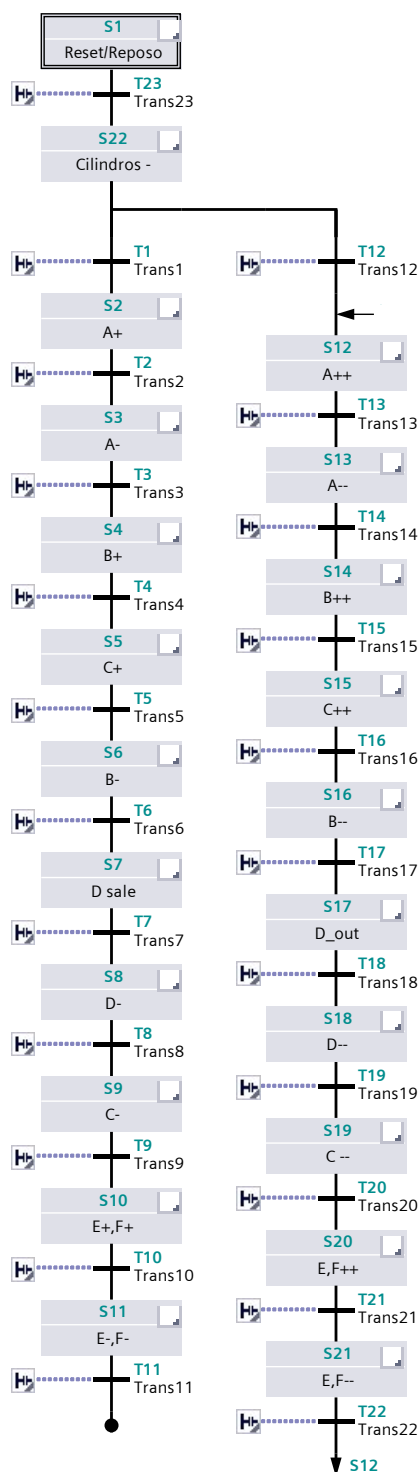
Enable alarms

True

Category	Category enabler	Display class
Error		0
Warning		0
Info		0
Category 4		0
Category 5		0
Category 6		0
Category 7		0
Category 8		0

Category for interlocks	Error	Subcategory 1 for interlocks	Subcategory 2 for interlocks

Totally Integrated Automation Portal					
Category for supervisions	Error	Subcategory 1 for supervisions		Subcategory 2 for supervisions	
Category for GRAPH warnings	Warning	Subcategory 1 for GRAPH warnings		Subcategory 2 for GRAPH warnings	
Symbol	Address	Type	Comment		
Permanent pre-instructions					
1:					
Sequences (1)					
1:Secuencia Cilindros					



S1 - [Initial step]:Reset/Reposo

Interlock -(c)-:

Interlock alarm

Alarm text

Totally Integrated Automation Portal		
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Interlock
(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision
(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		N	"reset"

T23:Trans23

%M3.7
"start"

S2:A+

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock
(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision
(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"CiI_A_+"

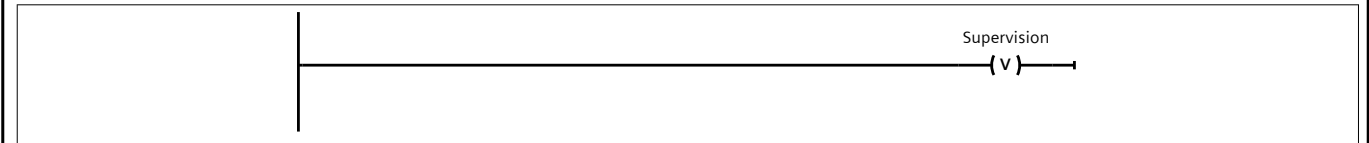
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Totally Integrated Automation Portal			
T2:Trans2			
%M1.5 "A+"			
S3:A-			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
Supervision (v)			
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		S	"CiI_A_-"
		R	"CiI_A_+"
T3:Trans3			
%M0.7 "A-"			
S4:B+			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			

Supervision -(v)-:

Supervision alarm

Alarm text	
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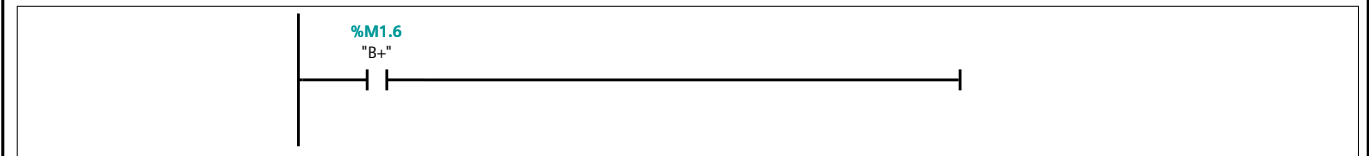


Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"Cil_B_Out"

T4:Trans4

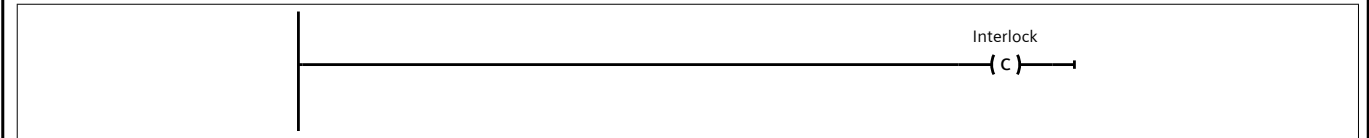


S5:C+

Interlock -(c)-:

Interlock alarm	
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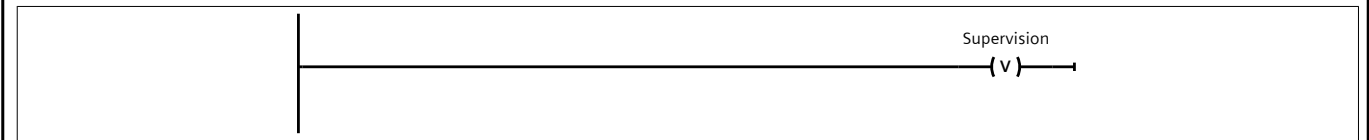
Alarm text	
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Supervision -(v)-:

Supervision alarm

Alarm text	
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Actions:

Actions:

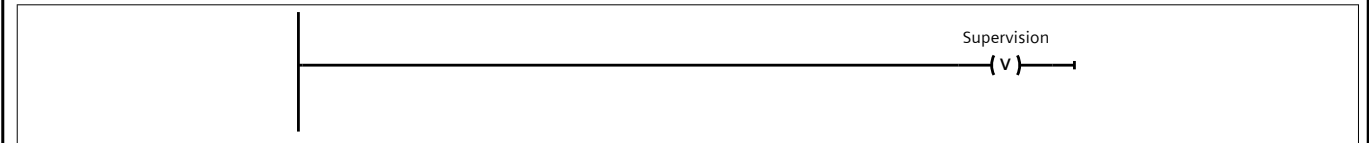
Interlock	Event	Qualifier	Action
		S	"Cil_C_Out"

Totally Integrated Automation Portal			
T5:Trans5			
%M1.7 "C+"			
S6:B-			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
Supervision (v)			
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		R	"CiI_B_Out"
T6:Trans6			
%M1.0 "B-"			
S7:D sale			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			

Supervision -(v)-:

Supervision alarm

Alarm text	
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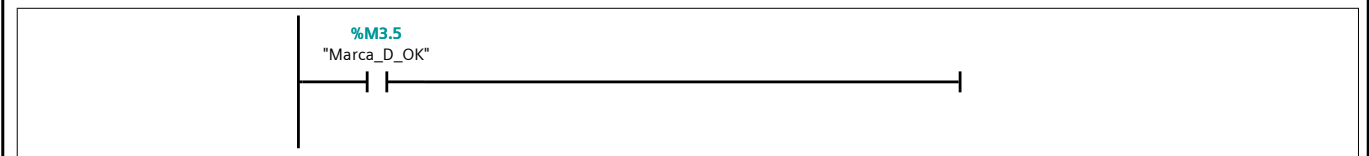


Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"Cil_D_Out"

T7:Trans7			

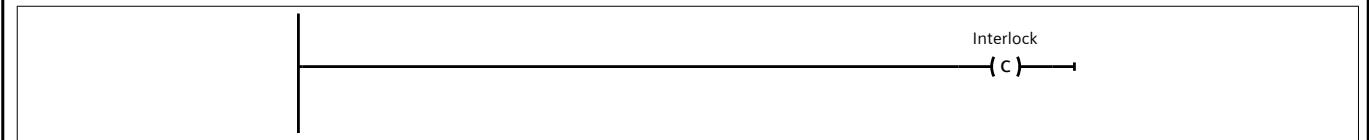


S8:D-

Interlock -(c)-:

Interlock alarm

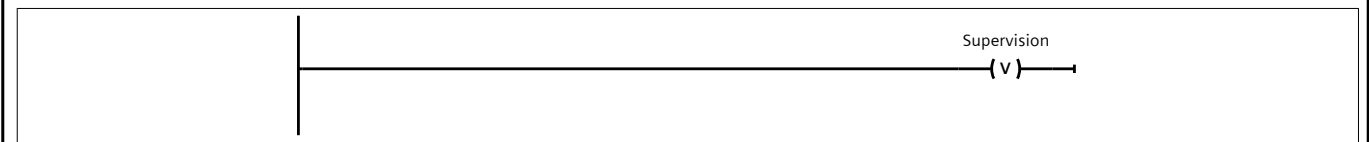
Alarm text	
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Supervision -(v)-:

Supervision alarm

Alarm text	
------------	--



Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"Cil_D_Out"

Totally Integrated Automation Portal		
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T8:Trans8

%M1.2

"D-"

S9:C-

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"CiI_C_Out"

T9:Trans9

%M1.1

"C-"

S10:E+,F+

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Totally Integrated Automation Portal			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
	Supervision (v)		
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		S	"CiI_E_+"
		S	"CiI_F_+"
T10:Trans10			
	%M2.1 "E+" %M2.2 "F+"		
S11:E-,F-			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
	Interlock (c)		
Supervision -(v)-:			
Supervision alarm			
Alarm text			
	Supervision (v)		
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		S	"CiI_E_-"
		S	"CiI_F_-"

Totally Integrated Automation Portal			
T11:Trans11			
%M2.1 "E+" %M2.2 "F+"			
S12:A++			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
Supervision (v)			
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		S	"CiI_A_+"
T13:Trans13			
%M1.5 "A+"			
S13:A--			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			

Totally Integrated Automation Portal		
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Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"CiI_A_-"
		R	"CiI_A_+"

T14:Trans14

%M0.7

"A-"

S14:B++

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"CiI_B_Out"

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Totally Integrated Automation Portal		
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T15:Trans15

%M1.6

"B+"

S15:C++

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

Alarm text

Supervision

(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"CiI_C_Out"

T16:Trans16

%M1.7

"C+"

S16:B--

Interlock -(c)-:

Interlock alarm

Alarm text

Interlock

(c)

Supervision -(v)-:

Supervision alarm

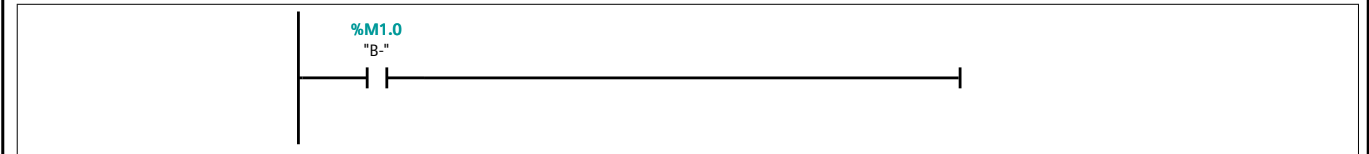
Alarm text	
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Actions:

Actions:

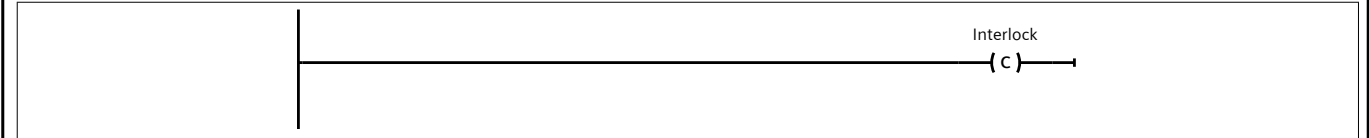
Interlock	Event	Qualifier	Action
		R	"Cil_B_Out"

[illegible]

S17:D_out

Interlock alarm

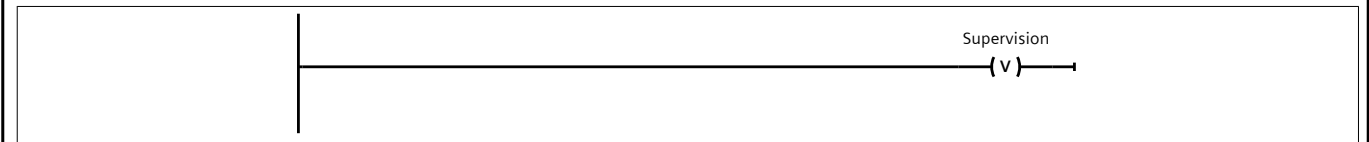
Alarm text	
------------	--



Supervision -(v)-:

Supervision alarm

Alarm text	
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Actions:

Actions:

Interlock	Event	Qualifier	Action
		S	"Cil_D_Out"

Totally Integrated Automation Portal			
T18:Trans18			
%M3.5 "Marca_D_OK"			
S18:D--			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
Supervision (v)			
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		R	"CiI_D_Out"
T19:Trans19			
%M1.2 "D-"			
S19:C --			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			

Totally Integrated Automation Portal			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
	Supervision (v)		
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		R	"CiI_C_Out"
T20:Trans20			
	%M1.1 "C-"		
S20:E,F++			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
	Interlock (c)		
Supervision -(v)-:			
Supervision alarm			
Alarm text			
	Supervision (v)		
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		S	"CiI_E_+"
		S	"CiI_F_+"

Totally Integrated Automation Portal			
T21:Trans21			
%M2.1 "E+" %M2.2 "F+"			
S21:E,F--			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			
Supervision -(v)-:			
Supervision alarm			
Alarm text			
Supervision (v)			
Actions:			
Actions:			
Interlock	Event	Qualifier	Action
		S	"CiI_E_-"
		S	"CiI_F_-"
T22:Trans22			
%M2.1 "E+" %M2.2 "F+"			
S22:Cilindros -			
Interlock -(c)-:			
Interlock alarm			
Alarm text			
Interlock (c)			

Totally Integrated Automation Portal		
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Supervision -(v)-:

Supervision alarm

Alarm text

Supervision
(v)

Actions:

Actions:

Interlock	Event	Qualifier	Action
		R	"CiI_A_-"
		R	"CiI_E_-"
		R	"CiI_F_-"

T1:Trans1

%M0.0
"Btn_Ciclo_INIT"

%M0.4
"Sens_PP"

%M0.5
"Sens_PC"

T12:Trans12

%M0.1
"Btn_Ciclo_Cont"

%M0.4
"Sens_PP"

%M0.5
"Sens_PC"

Permanent post-instructions

1: